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DATE: 7.2.26

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EXPERIMENT 4

ANALYZE THE MALICIOUS NETWORK TRAFFIC USING WIRESHARK

AIM:

To analyse captured network traffic using Wireshark and identify malicious activities.

ALGORITHM:

1. Open the PCAP file in Wireshark.
2. Apply basic filters (e.g., http, dns, tcp).
3. Identify suspicious IP addresses and domain.
4. Inspect unusual protocols or large data transfers.
5. Follow TCP streams to view communication.
6. Detect indicators of compromise (malware, phishing, etc).
7. Note down findings and analyze attack behavior.

Task 1 Scenario



▶ Start Machine



Eric Fischer from the Purchasing Department at Bartell Ltd has received an email from a known contact with a Word document attachment. Upon opening the document, he accidentally clicked on "Enable Content." The SOC Department immediately received an alert from the endpoint agent that Eric's workstation was making suspicious connections outbound. The pcap was retrieved from the network sensor and handed to you for analysis.

Task: Investigate the packet capture and uncover the malicious activities.

*Credit goes to Brad Duncan for capturing the traffic and sharing the pcap packet capture with infoSec community.

NOTE: DO NOT directly interact with any domains and IP addresses in this challenge.

Deploy the machine attached to this task; it will be visible in the **split-screen** view once it is ready.

If you don't see a virtual machine load, then click the **Show Split View** button.



Carnage

Apply your analytical skills to analyze the malicious network traffic using Wireshark.

[Start AttackBox](#) [Show Split View](#)

Answer the questions below

Read the above.

No answer needed

✓ Correct Answer

Task 2 Traffic Analysis



Are you ready for the journey?

Please, load the pcap file in your Analysis folder on the Desktop into Wireshark to answer the questions below.

Answer the questions below

What was the date and time for the first HTTP connection to the malicious IP?

(answer format: yyyy mm dd hh:mm:ss)

2021 09 24 16:44:38

✓ Correct Answer

What is the name of the zip file that was downloaded?

documents.zip

✓ Correct Answer

What was the domain hosting the malicious zip file?

attirenepal.com

✓ Correct Answer

Without downloading the file, what is the name of the file in the zip file?

chart 1530076591.xls

✓ Correct Answer

What is the name of the webserver of the malicious IP from which the zip file was downloaded?

LiteSpeed

✓ Correct Answer

What is the version of the webserver from the previous question?

PHP/7.2.34

✓ Correct Answer

Malicious files were downloaded to the victim host from multiple domains. What were the three domains involved with this activity?

finejewels.com.au, thietbiagt.com, new.americold.com

✓ Correct Answer



Which certificate authority issued the SSL certificate to the first domain from the previous question?

GoDaddy

✓ Correct Answer

What are the two IP addresses of the Cobalt Strike servers? Use VirusTotal (the Community tab) to confirm if IPs are identified as Cobalt Strike C2 servers. (answer format: enter the IP addresses in sequential order)

185.106.96.158, 185.125.204.174

✓ Correct Answer



What is the Host header for the first Cobalt Strike IP address from the previous question?

ocsp.verisign.com

✓ Correct Answer

What is the domain name for the first IP address of the Cobalt Strike server? You may use VirusTotal to confirm if it's the Cobalt Strike server (check the Community tab).

survmeter.live

✓ Correct Answer



What is the domain name of the second Cobalt Strike server IP? You may use VirusTotal to confirm if it's the Cobalt Strike server (check the Community tab).

securitybusinpuuff.com

✓ Correct Answer



What is the domain name of the post infection traffic?

maldivehost.net

✓ Correct Answer



What are the first eleven characters that the victim host sends out to the malicious domain involved in the post infection traffic?

zLisQRWZl9

✓ Correct Answer

What was the length for the first packet sent out to the C2 server?

281

✓ Correct Answer

What was the Server header for the malicious domain from the previous question?

Apache/2.4.49 (cPanel) OpenSSL/1.1.1l mod_bwlimited/1.4

✓ Correct Answer

The malware used an API to check for the IP address of the victim's machine. What was the date and time when the DNS query for the IP check domain occurred? (answer format: yyyy-mm-dd hh:mm:ss UTC)

2021-09-24 17:00:04

✓ Correct Answer

What was the domain in the DNS query from the previous question?

api.ipify.org

✓ Correct Answer

Looks like there was some malicious spam (malspam) activity going on. What was the first MAIL FROM address observed in the traffic?

farshin@mailfa.com

✓ Correct Answer

How many packets were observed for the SMTP traffic?

1439

✓ Correct Answer

RESULT:

Thus, network traffic was analysed to extract key malicious activity indicators and understand the attack behaviour.